# USCD Embedded C Assignment 2

By
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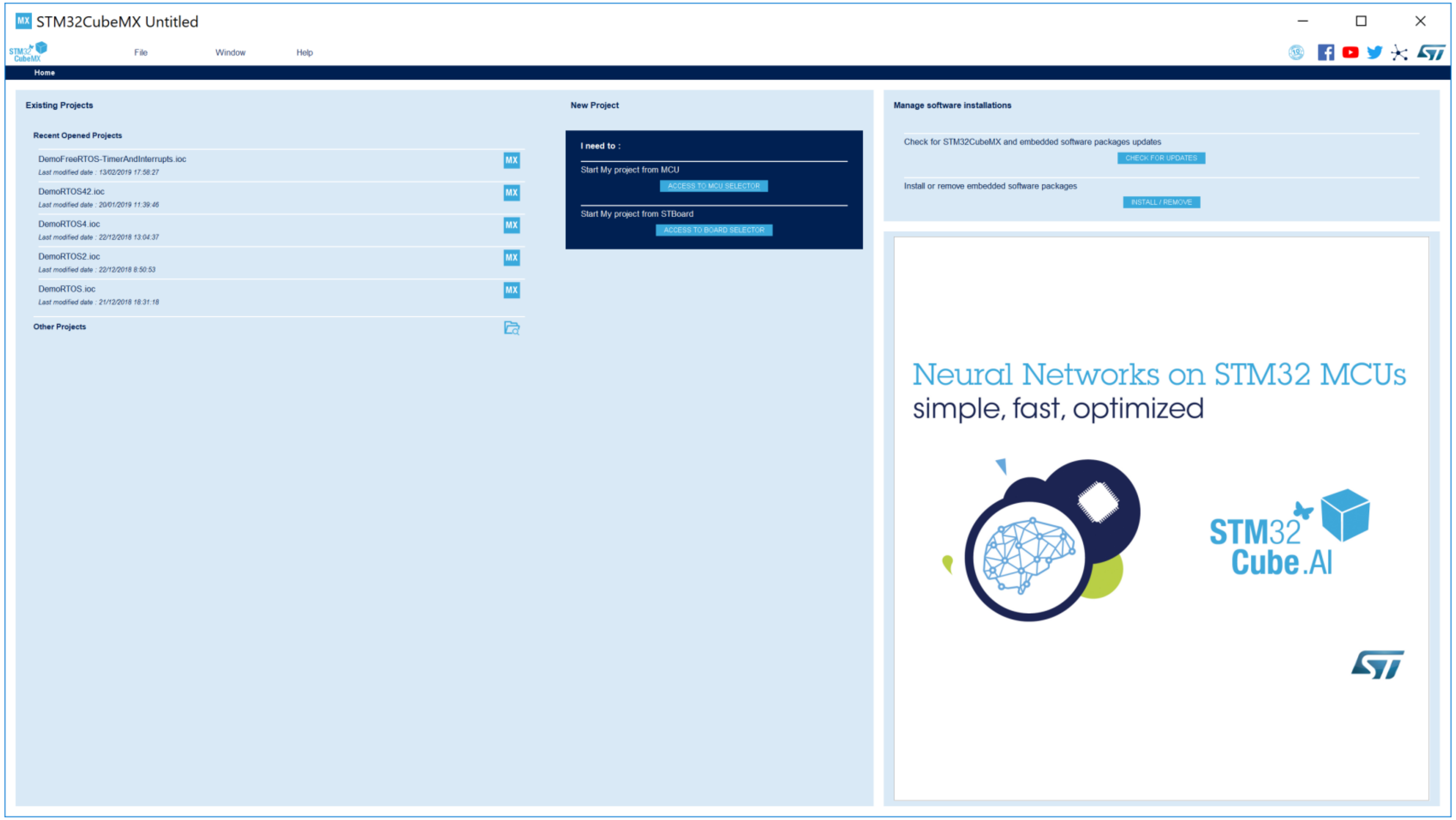
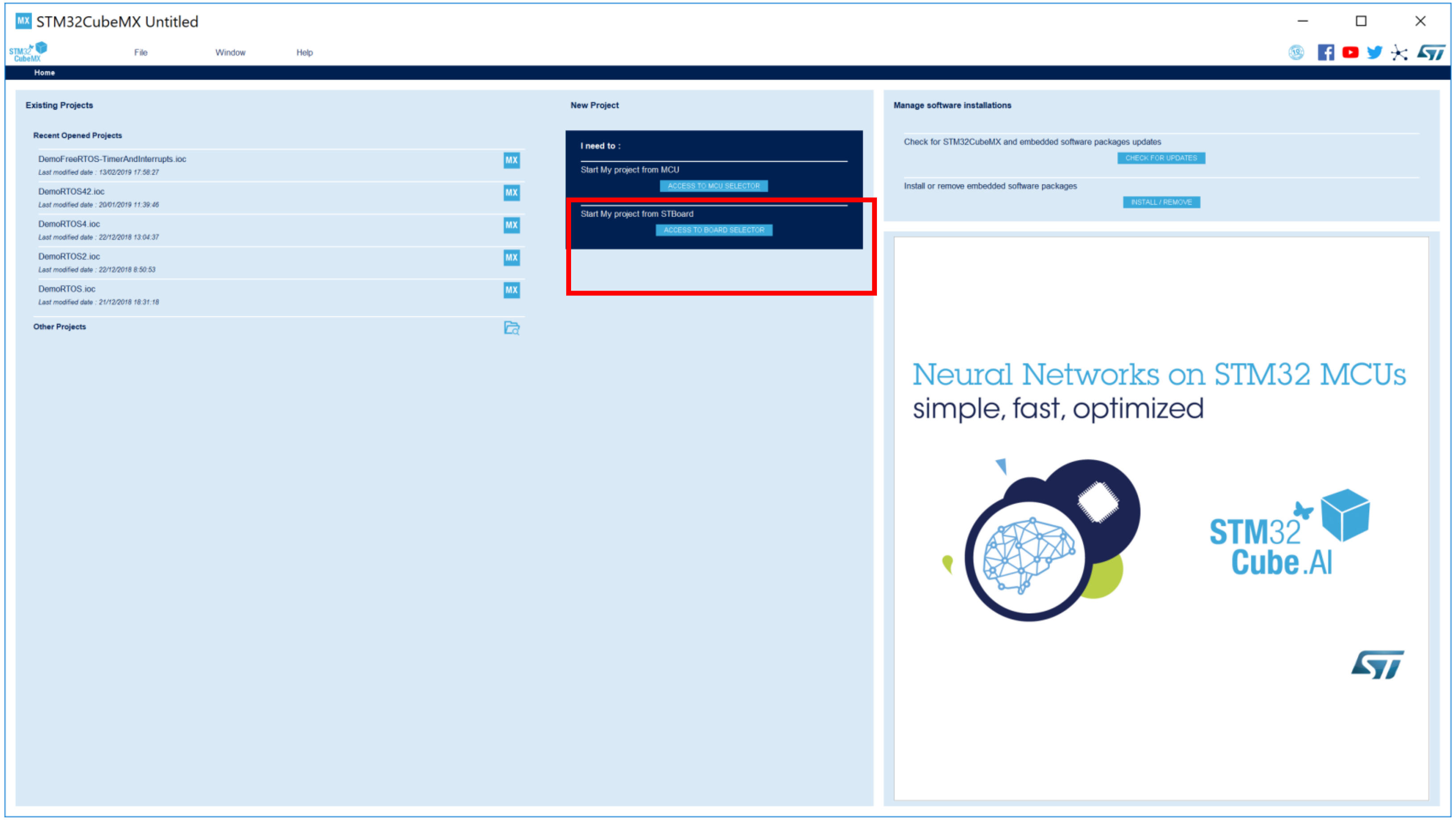
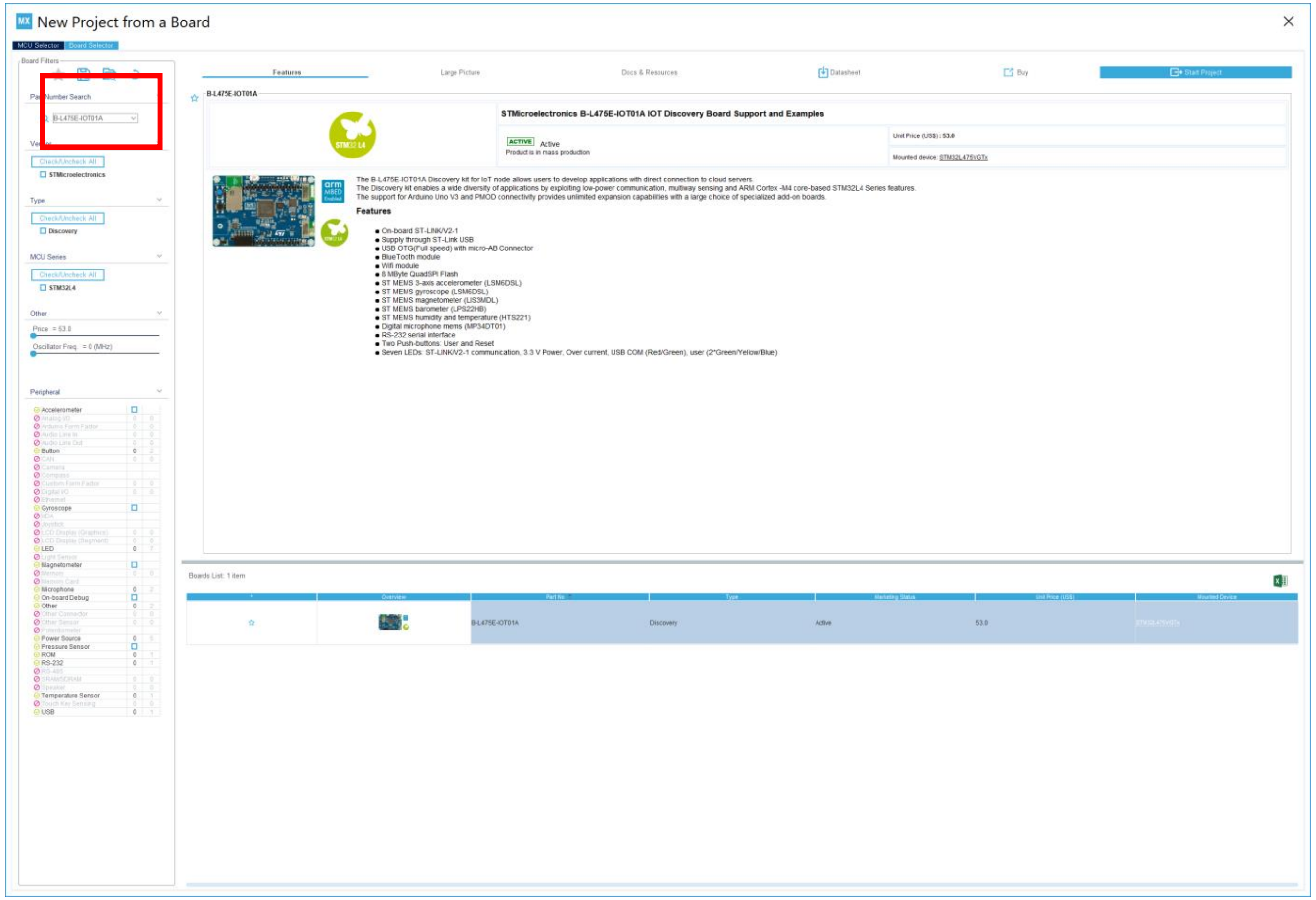
# Step 1. Startup STM32CubeMX



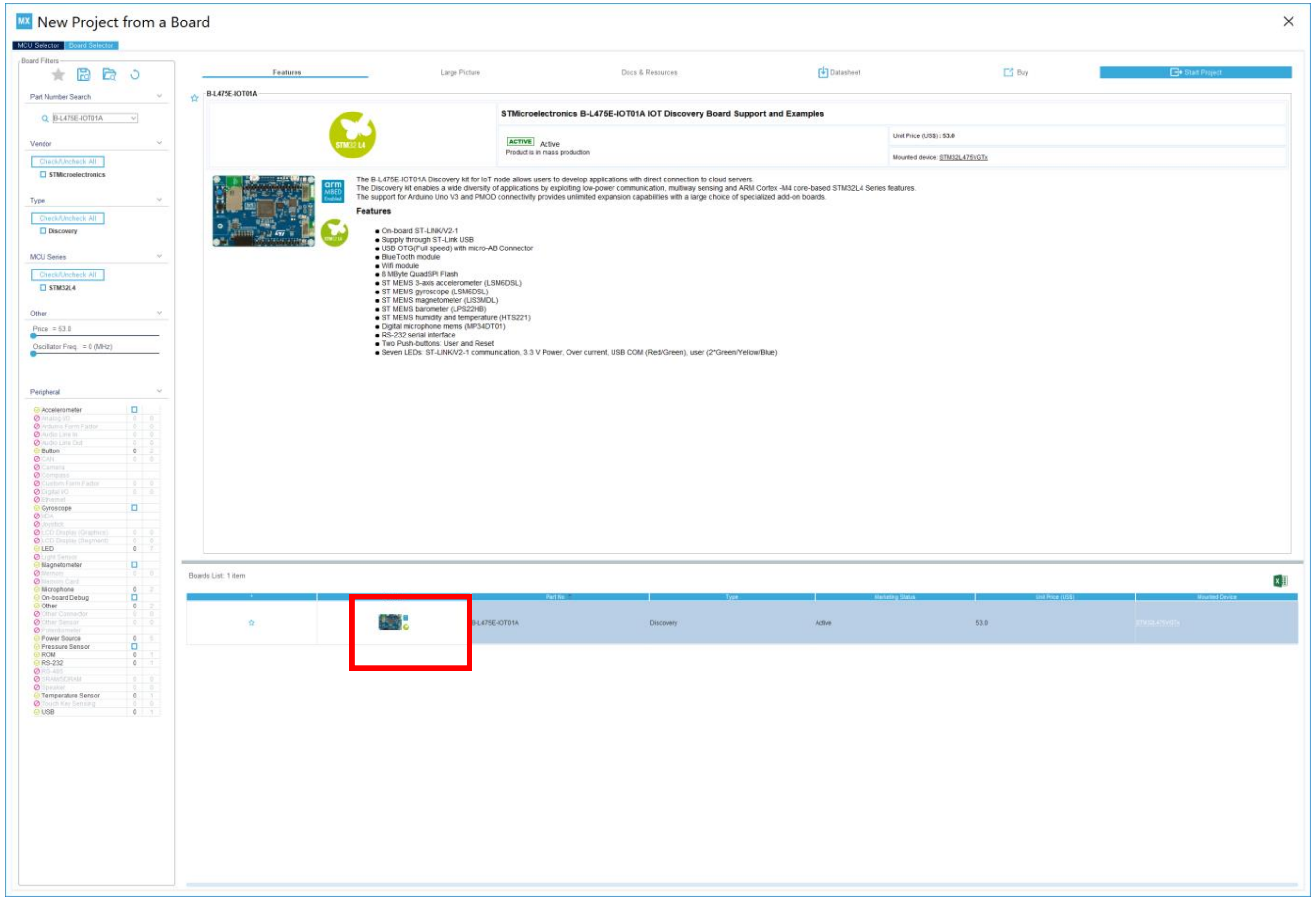
# Step 2. Access Board Selector



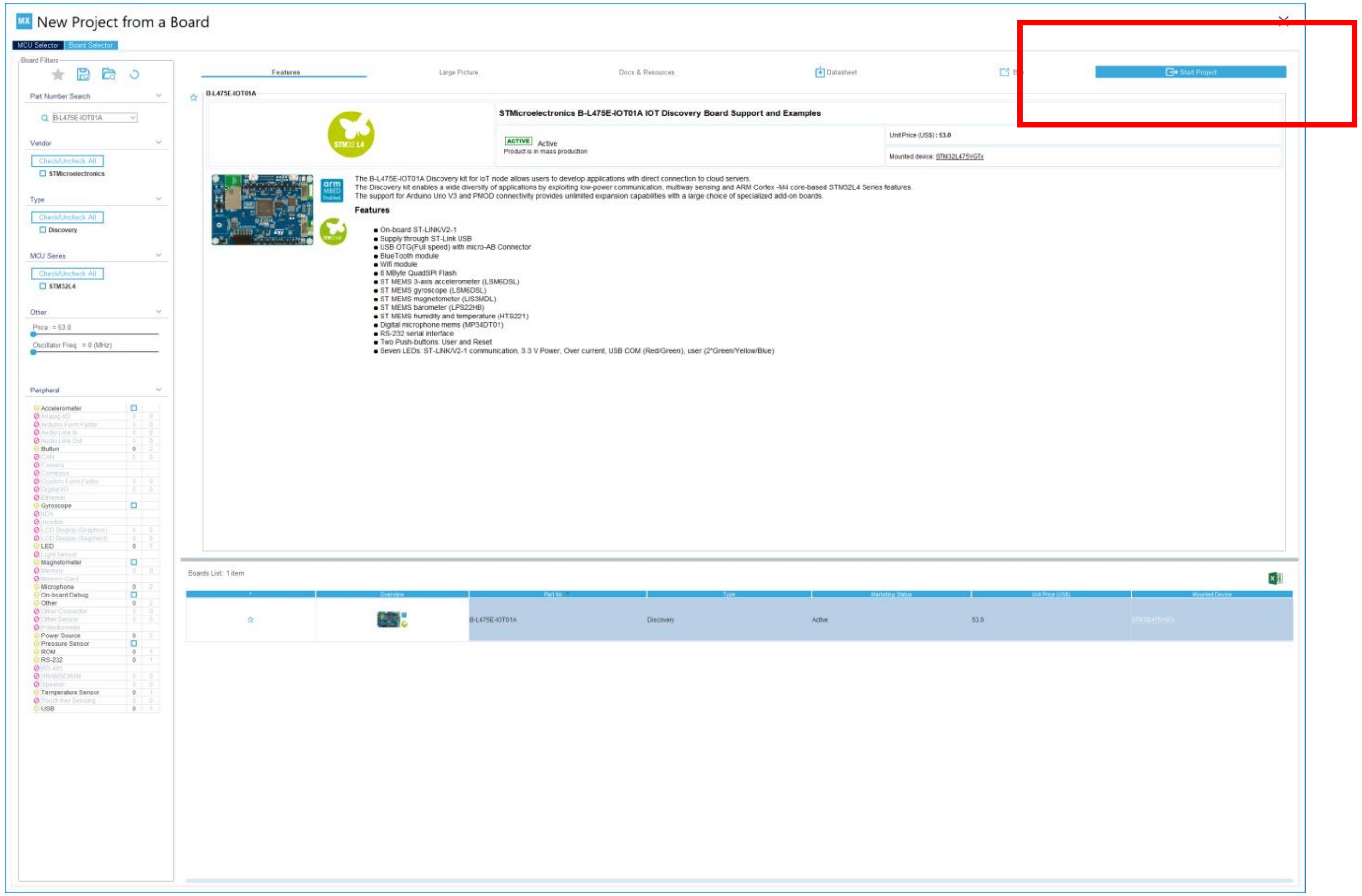
# Step 3. Enter “B-L475E-IOT01A” Board



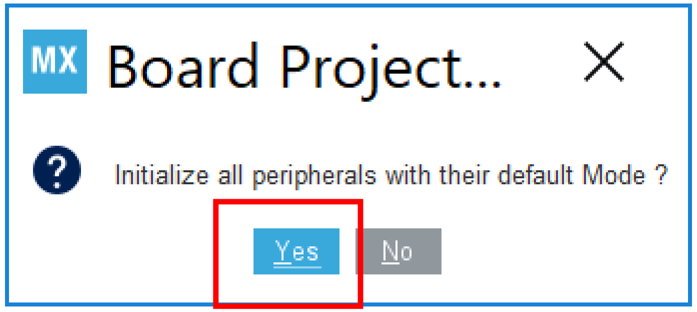
# Step 4. Select Board Photo



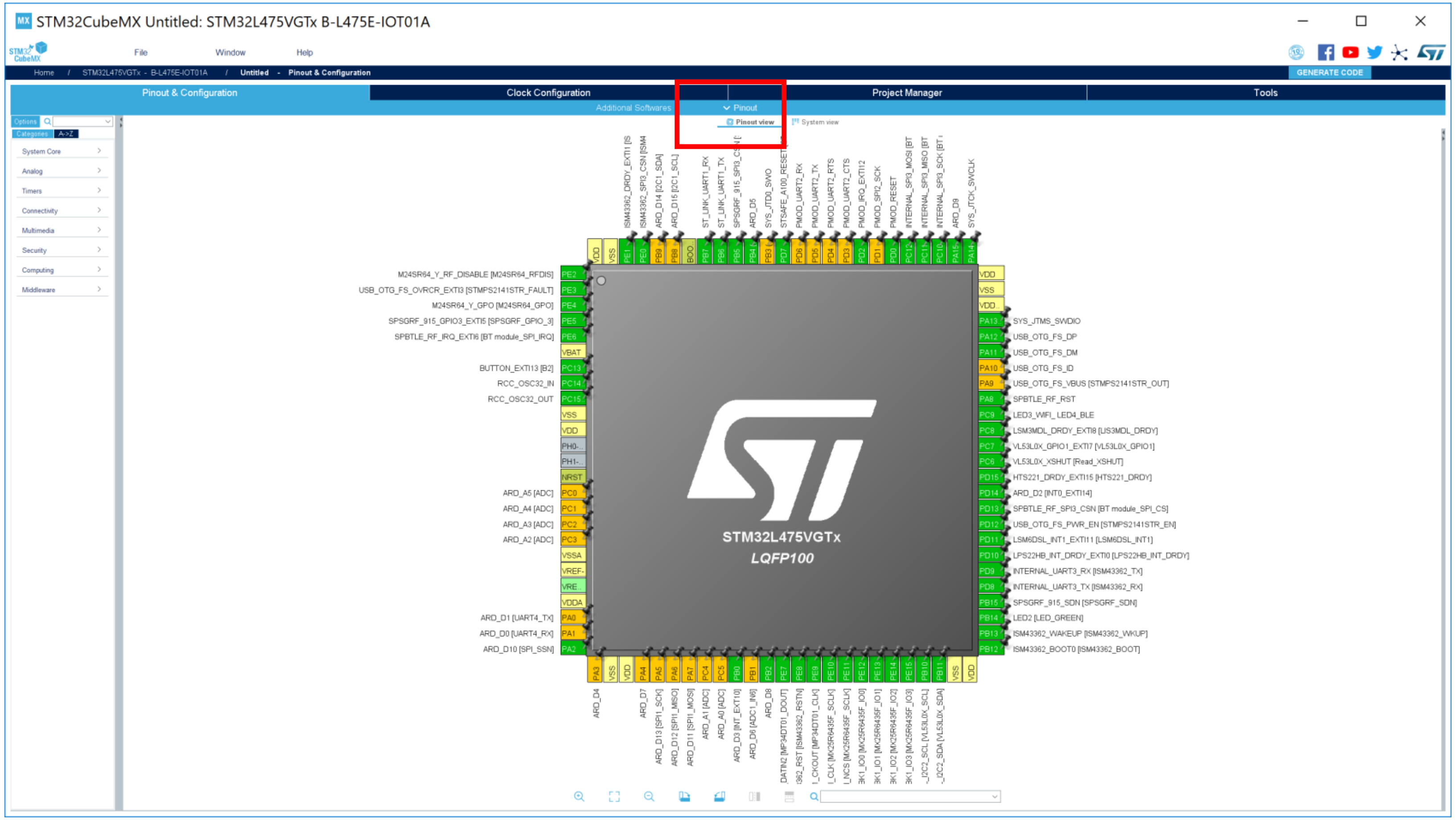
# Step 5. Select “Start Project”



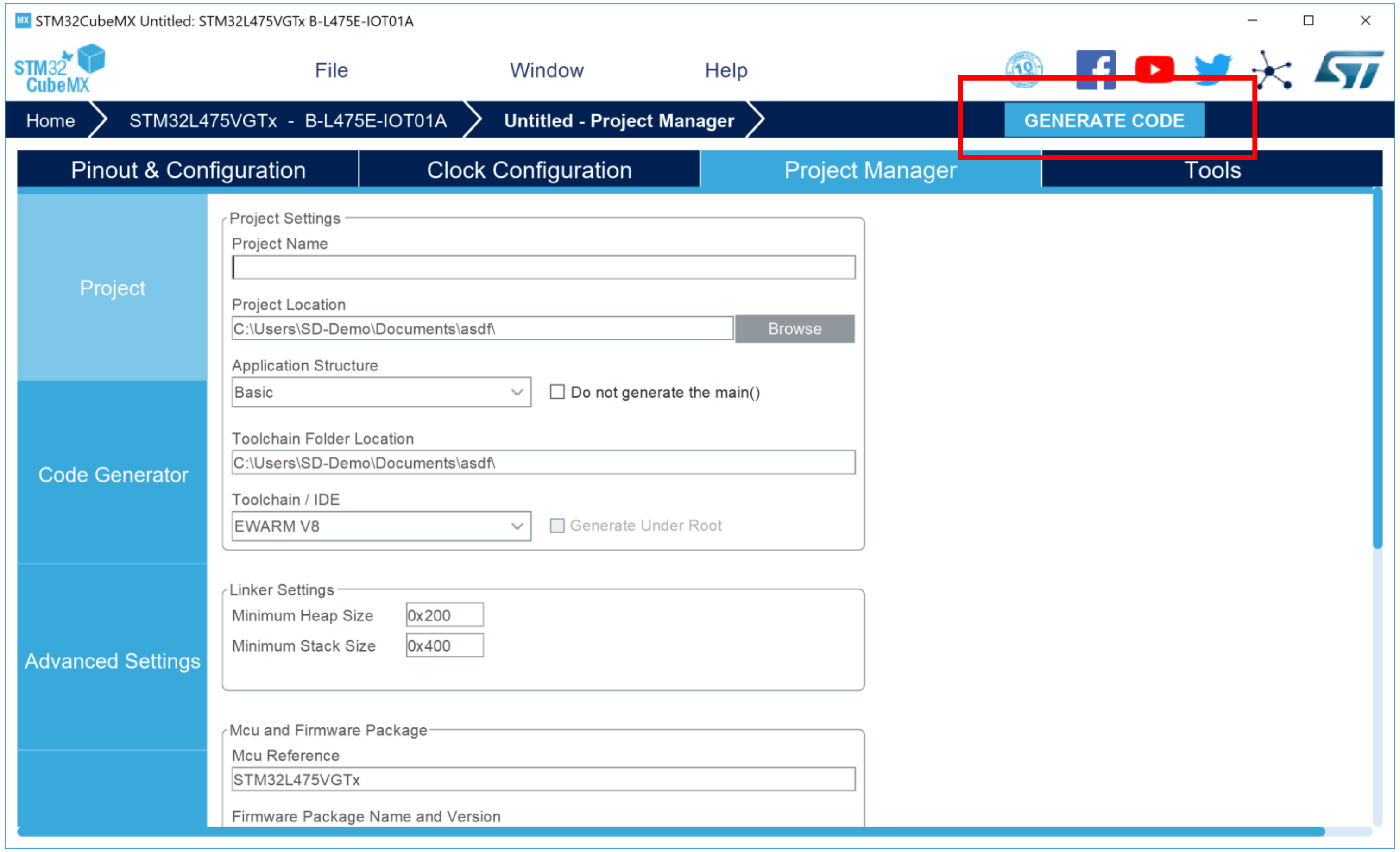
# Step 6. Select **YES** (initialize all peripherals with the default mode)



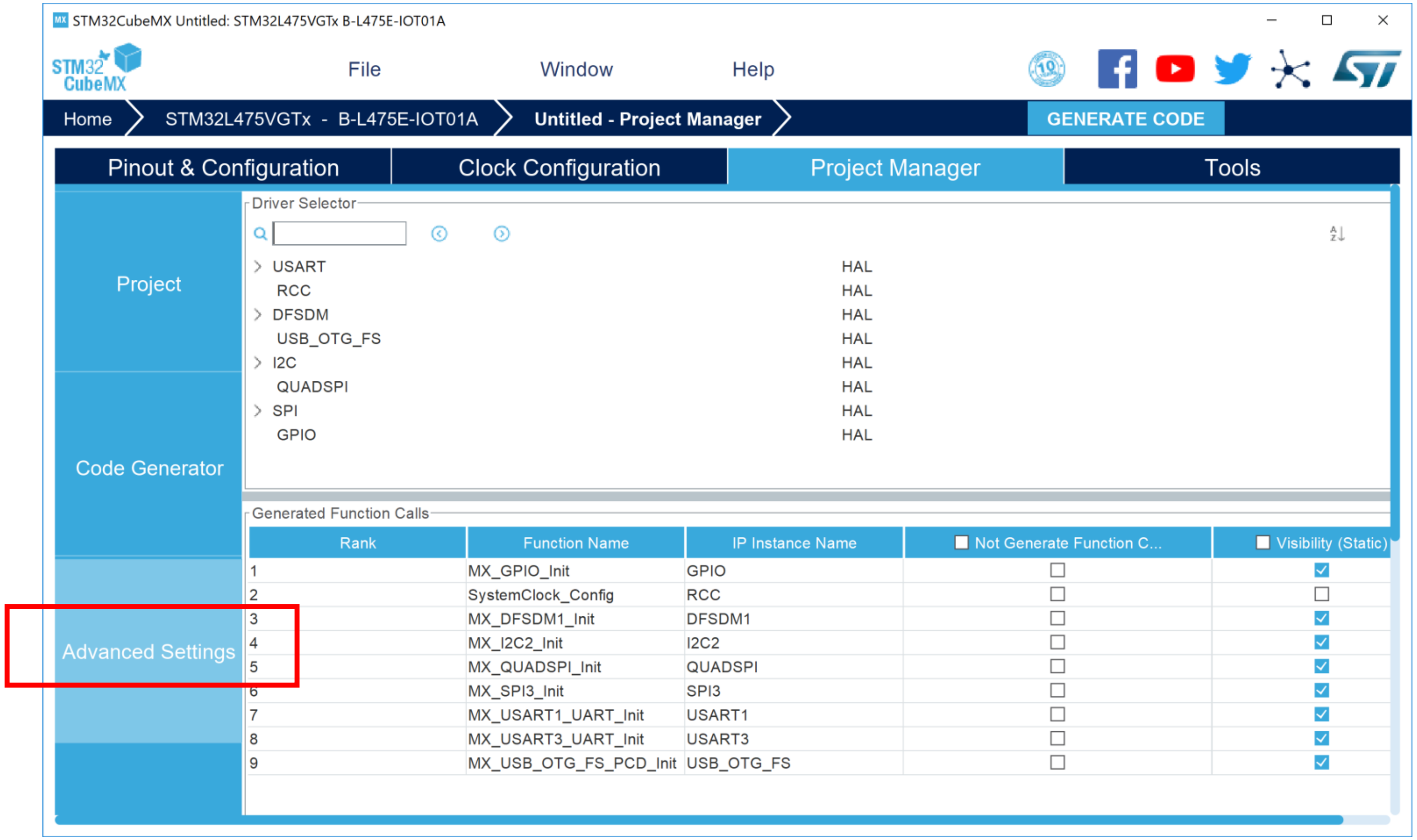
# Step 7. Observe Results (Pinout View)



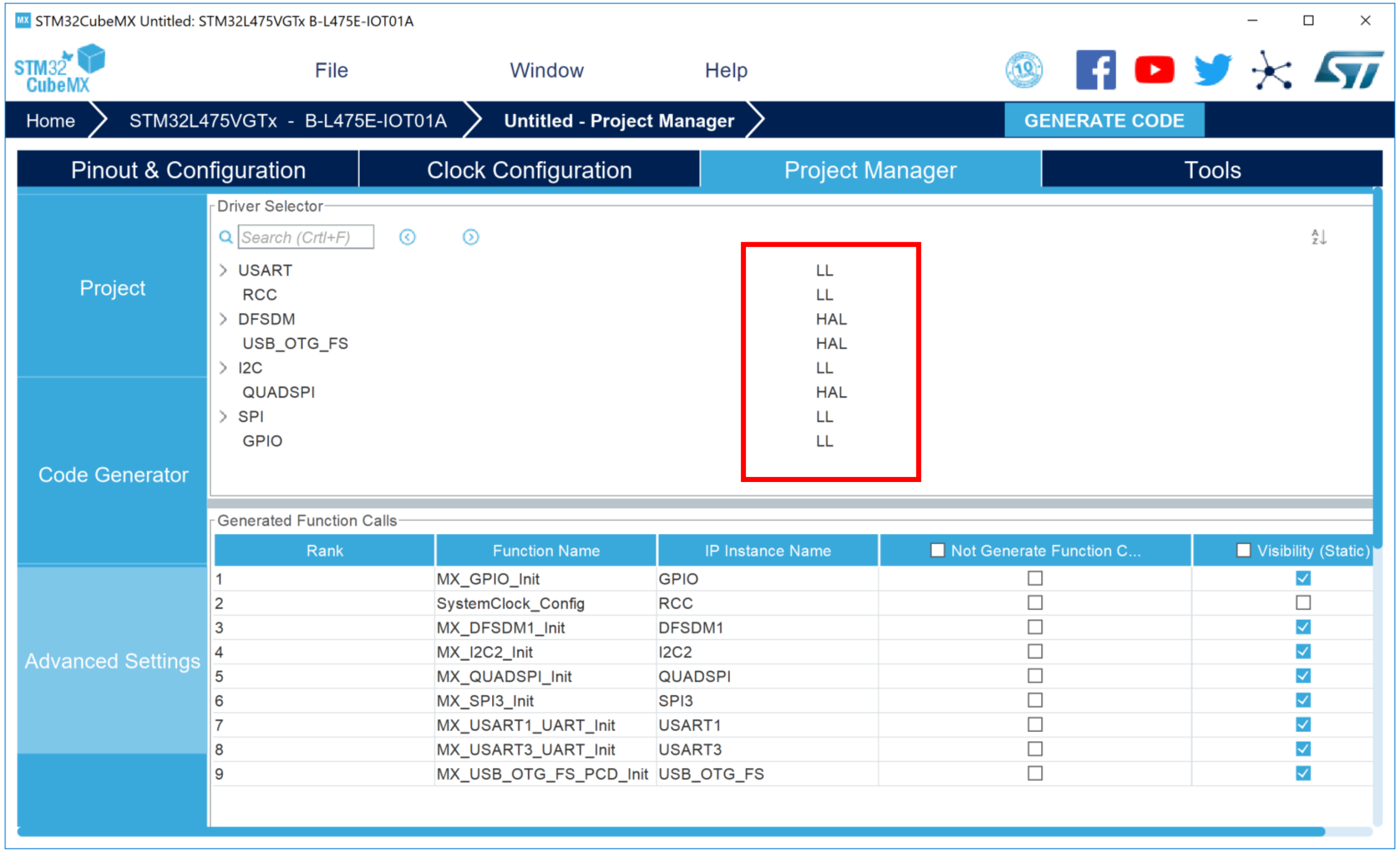
# Step 8. Select Project Manager Tab



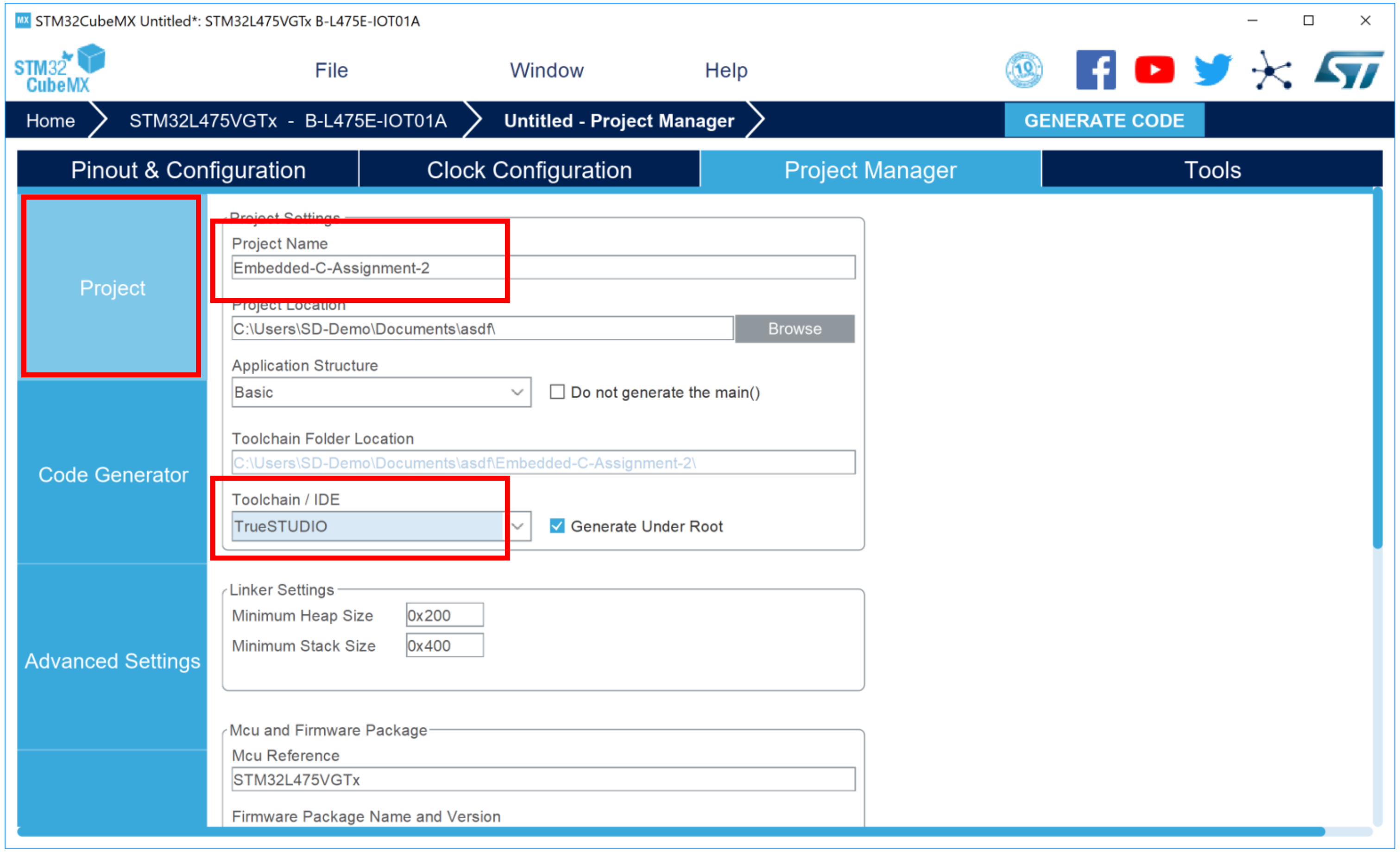
# Step 9. Select Advanced Settings



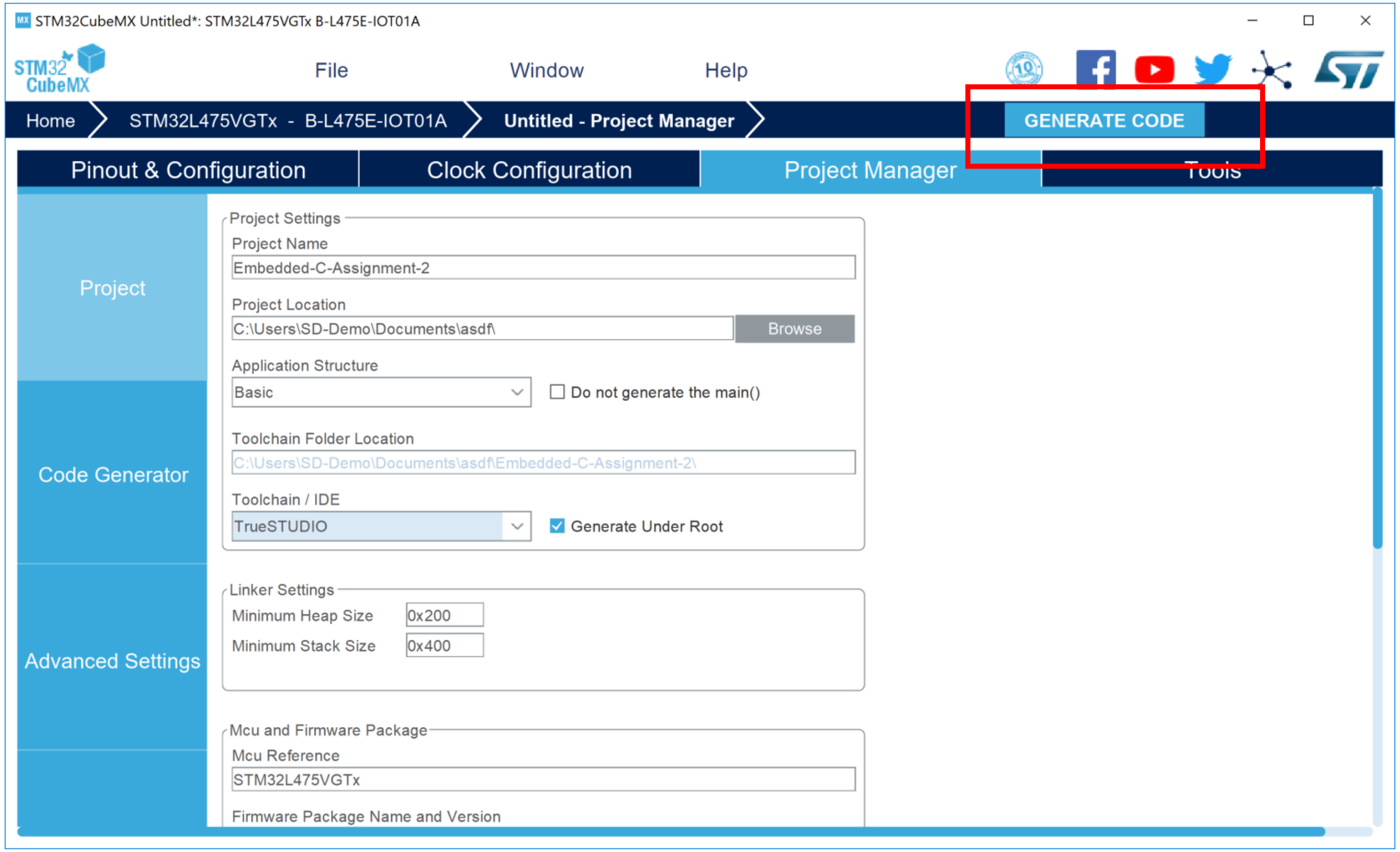
# Step 10. For Driver Selector, select LL for all that are available with LL option



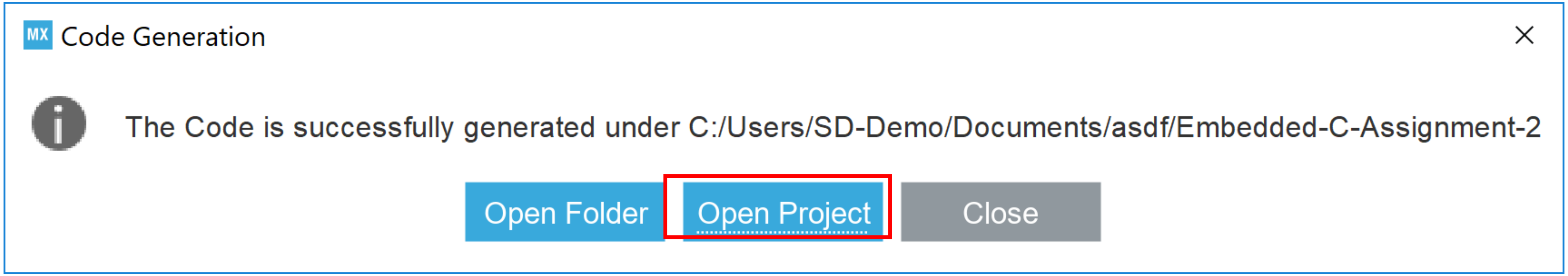
# Step 11. Enter “Embedded-C-Assigment-2” and select TrueStudio as IDE



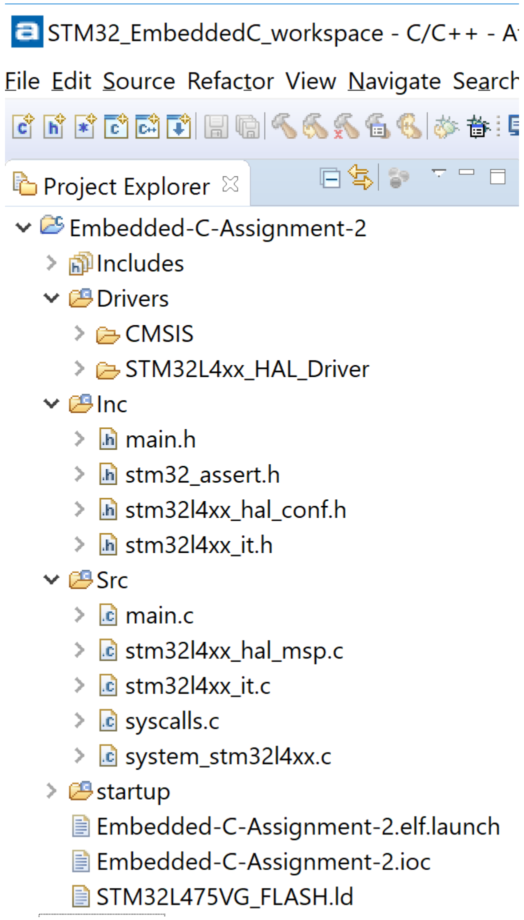
# Step 12. Select “Generate Code”



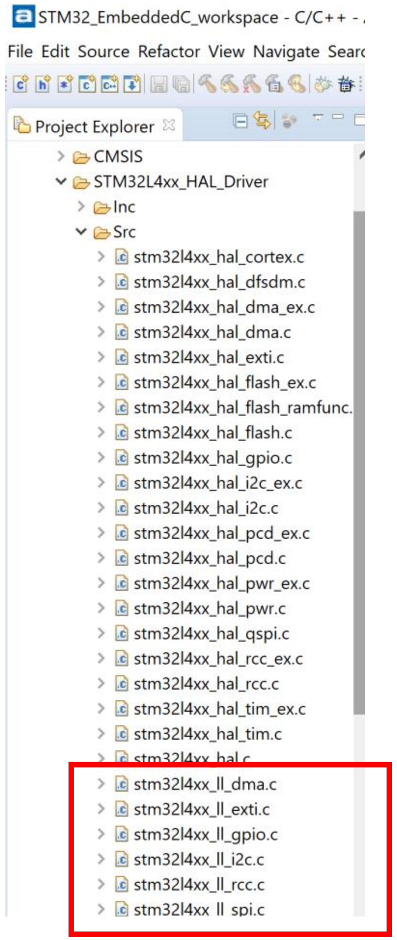
# Step 13. Select “Open Project”



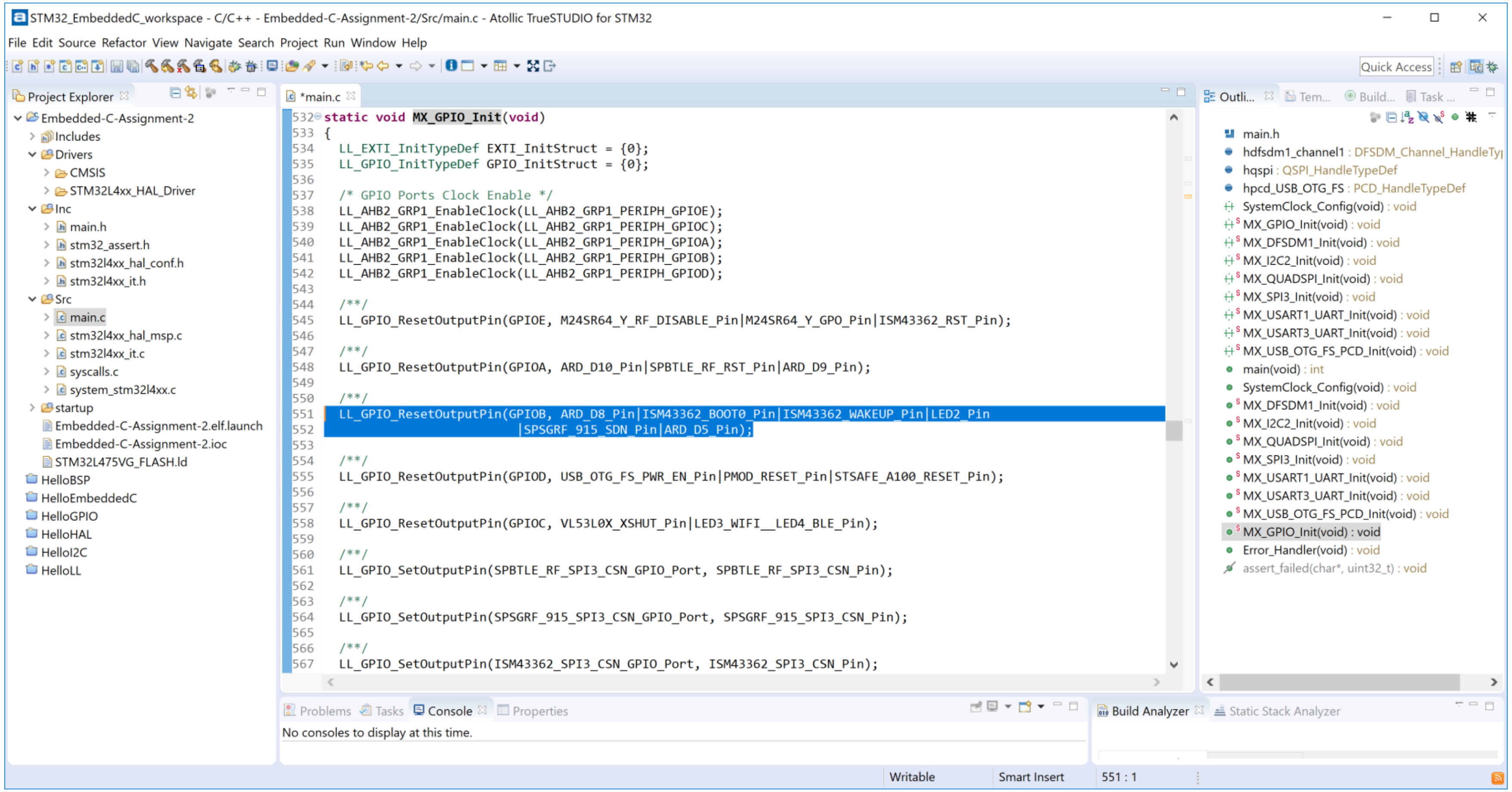
# Step 14. Resulting Project



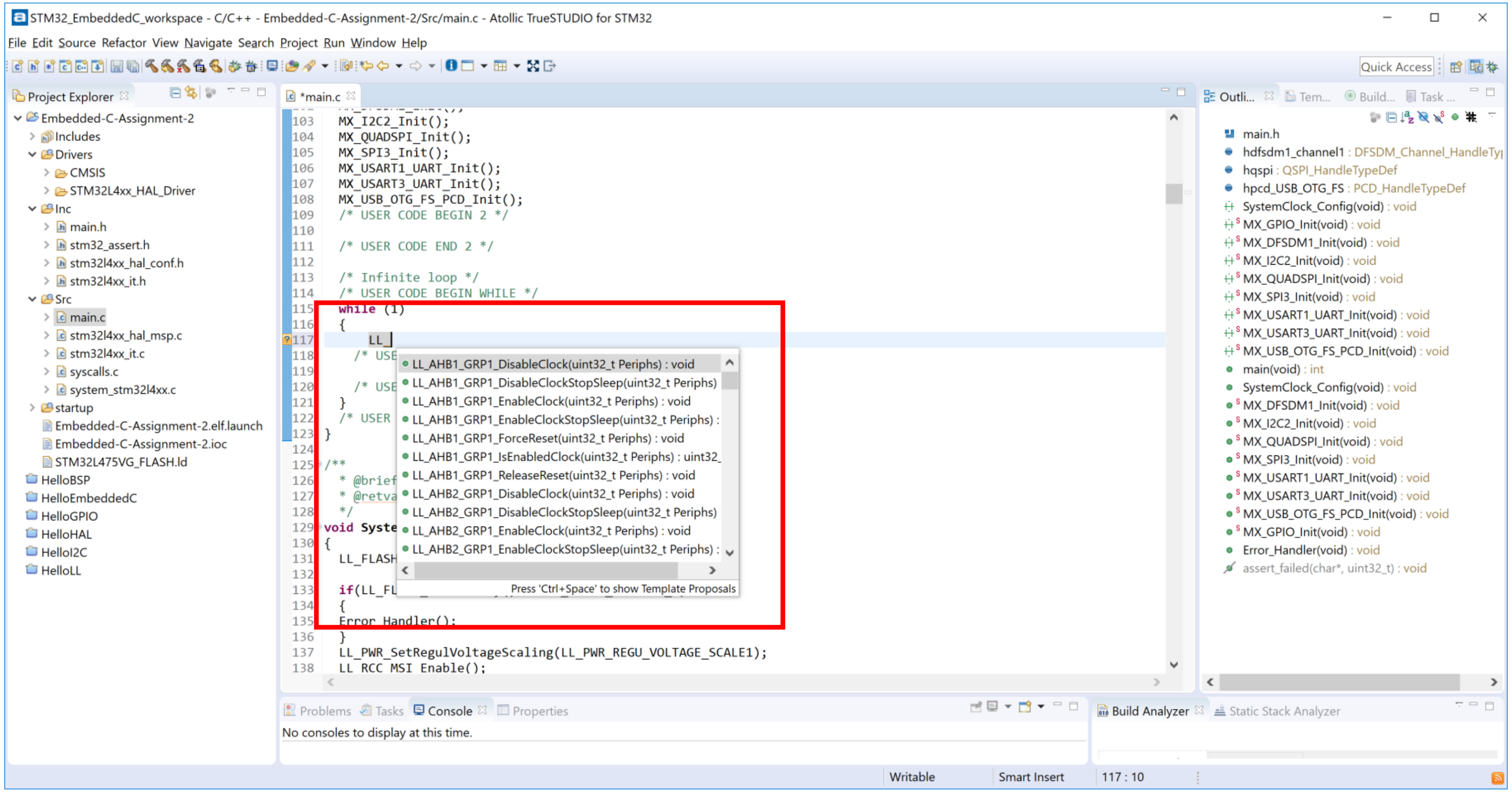
# Step 15. Confirm that LL drivers added to project



# Step 16. Find GPIO Init code that initializes the LED2 using LL (highlighted in blue below)



# Step 17. In main.c, inside the "while(1)" loop, enter "LL_" then press Ctrl+SpaceBar to observe LL_ APIs

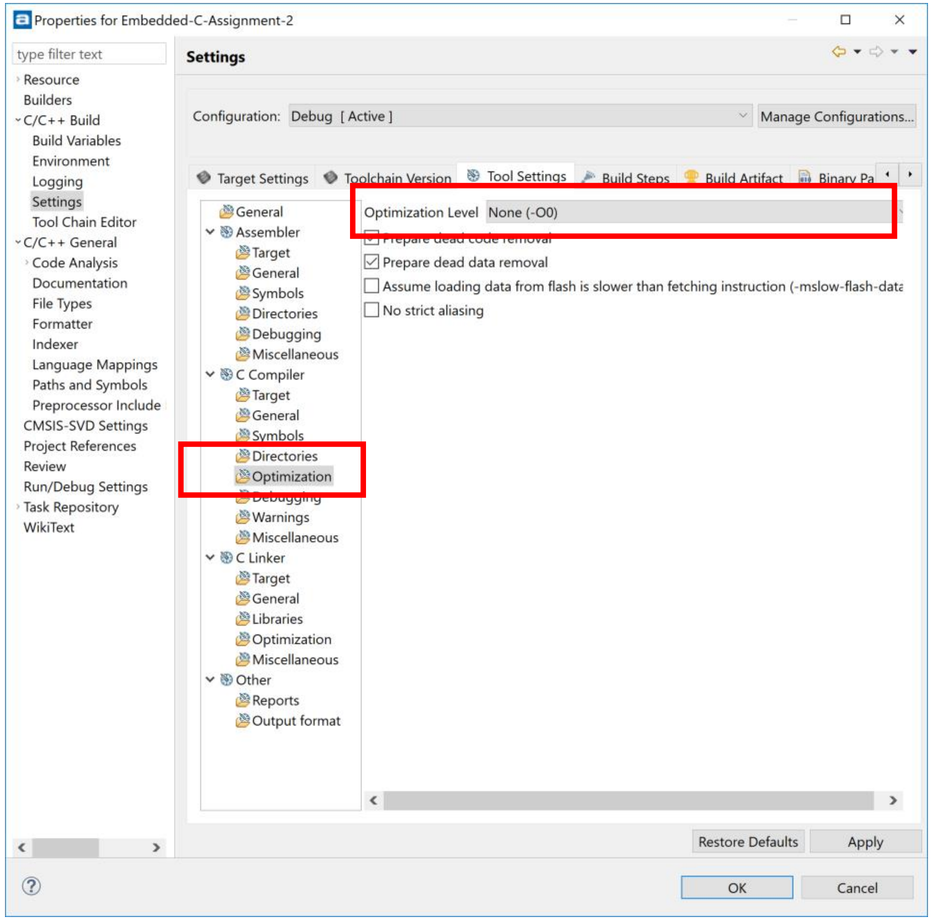


# Step 18. In main.c, enter LL related code shown below

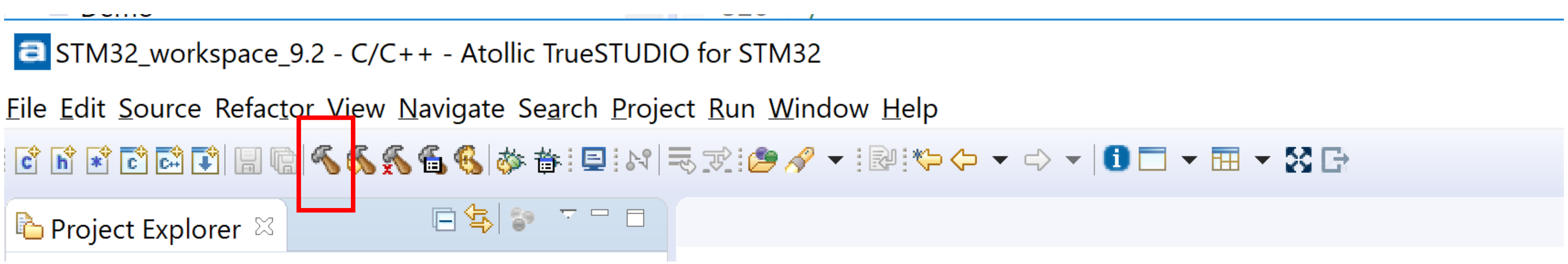
main.c

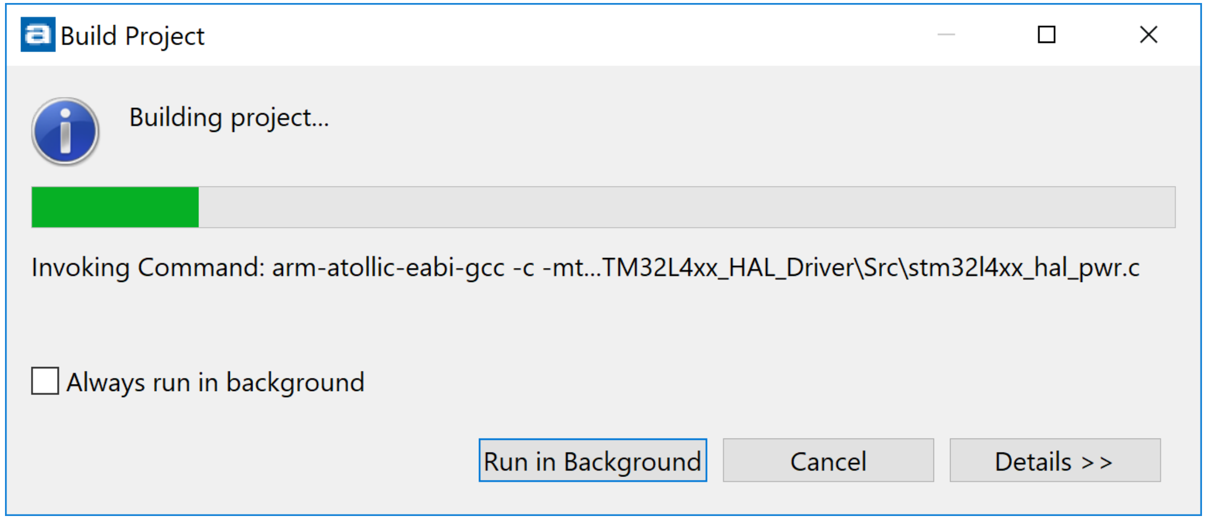
```c
  /* USER CODE END 2 */

  /* Infinite loop */
  /* USER CODE BEGIN WHILE */

  uint32_t flash_size = LL_GetFlashSize();
  printf("flash_size: %lx\n", flash_size);

  uint32_t uid[3];
  uid[0] = LL_GetUID_Word0();
  uid[1] = LL_GetUID_Word1();
  uid[2] = LL_GetUID_Word2();
  printf("uid: %lx %lx %lx\n", uid[0], uid[1], uid[2]);

  while (1)
  {
      LL_GPIO_TogglePin(GPIOB, LED2_Pin);

    LL_mDelay(1000);

    /* USER CODE END WHILE */

    /* USER CODE BEGIN 3 */
  }
```

# Step 19. Properties, C/C++ Build, Settings, C Compiler, Optimization, None



# Step 20. Build Project



# Step 21. Results of Build – Part 1

Problems   Tasks   Console   Properties

CDT Build Console [Embedded-C-Assignment-2]

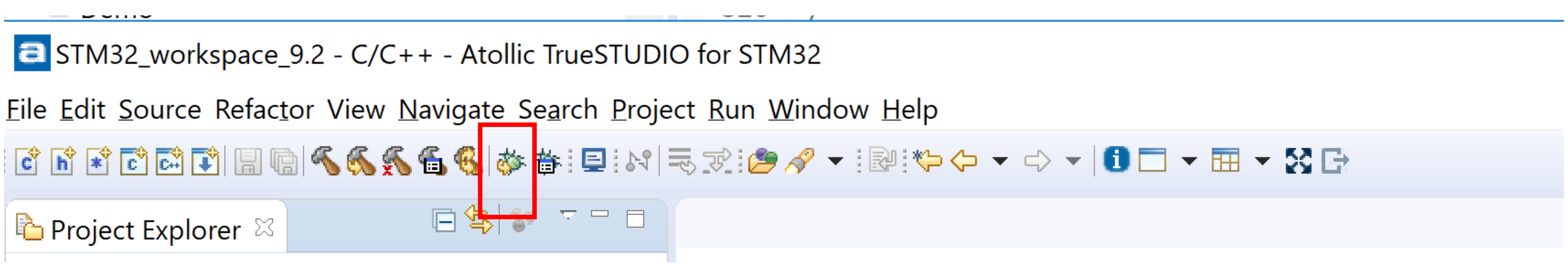
```
arm-atollic-eabi-gcc -c -mthumb -mcpu=cortex-m4 -mfloat-abi=hard -mfpu=fpv4-sp-d16 -std=gnu11 -DUSE_FULL_L
arm-atollic-eabi-gcc -c -mthumb -mcpu=cortex-m4 -mfloat-abi=hard -mfpu=fpv4-sp-d16 -std=gnu11 -DUSE_FULL_L
arm-atollic-eabi-gcc -c -mthumb -mcpu=cortex-m4 -mfloat-abi=hard -mfpu=fpv4-sp-d16 -std=gnu11 -DUSE_FULL_L
arm-atollic-eabi-gcc -o Embedded-C-Assignment-2.elf Drivers\STM32L4xx_HAL_Driver\Src\stm32l4xx_hal.o Drive
C:\Program Files (x86)\Atollic\TrueSTUDIO for STM32 9.2.0\ide\jre\bin\java -jar C:\Program Files (x86)\Ato
Generate build reports...
Print size information
   text    data     bss     dec     hex filename
  16636      12    2764   19412    4bd4 Embedded-C-Assignment-2.elf
Print size information done
Generate listing file
Output sent to: Embedded-C-Assignment-2.list
Generate listing file done
Generate build reports done
arm-atollic-eabi-objcopy.exe -O ihex Embedded-C-Assignment-2.elf Embedded-C-Assignment-2.hex

08:19:22 Build Finished (took 12s.503ms)
```

# Step 22. Results of Build – Part 2

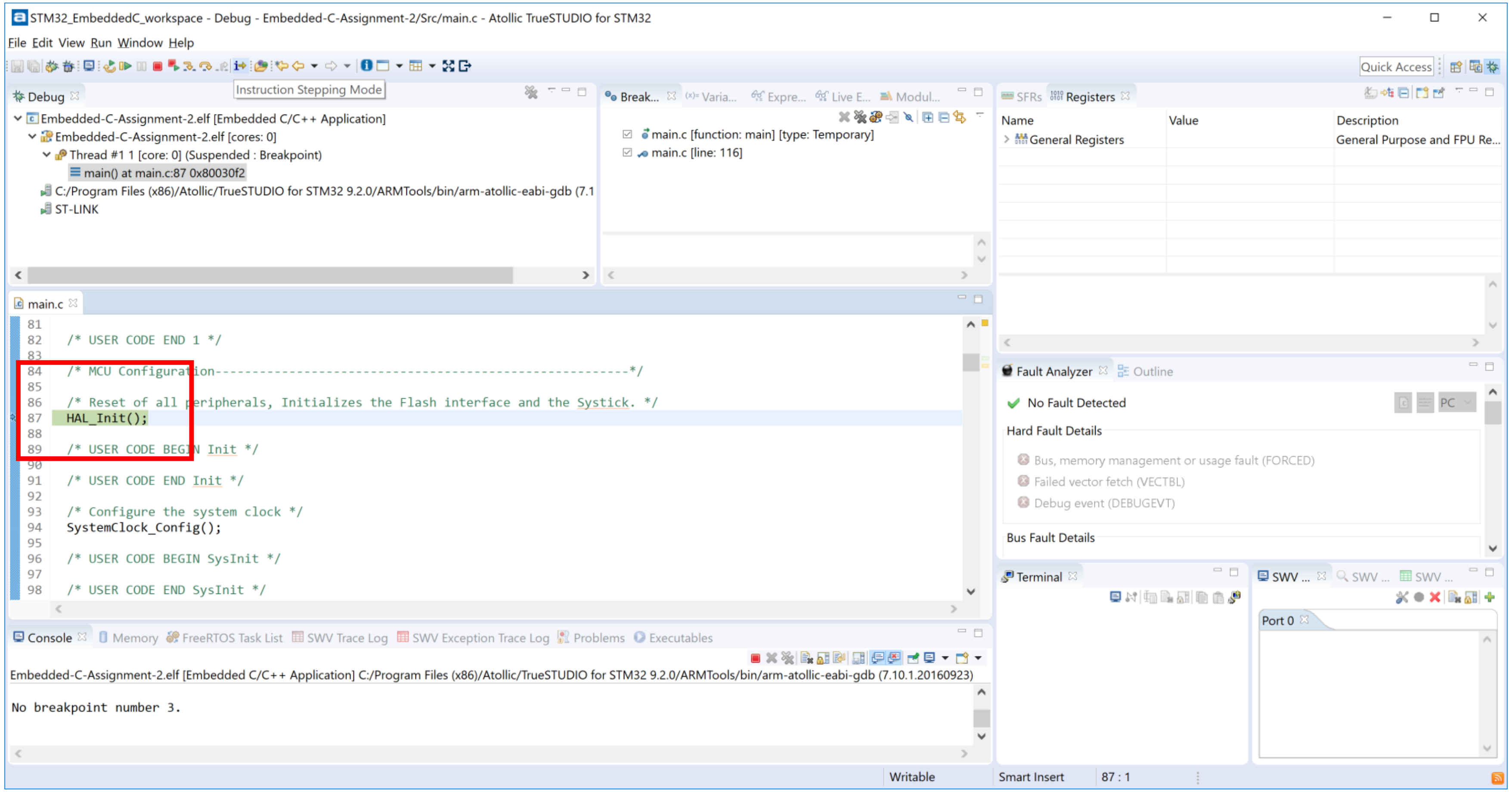
Memory Regions | Memory Details

| Region | Start address | End address | Size | Free | Used | Usage (%) |
|---|---|---|---|---|---|---|
| RAM | 0x20000000 | 0x20018000 | 96 KB | 93.18 KB | 2.82 KB | 2.93% |
| RAM2 | 0x10000000 | 0x10008000 | 32 KB | 32 KB | 0 B | 0.00% |
| FLASH | 0x08000000 | 0x08100000 | 1024 KB | 1003.48 KB | 20.52 KB | 2.00% |

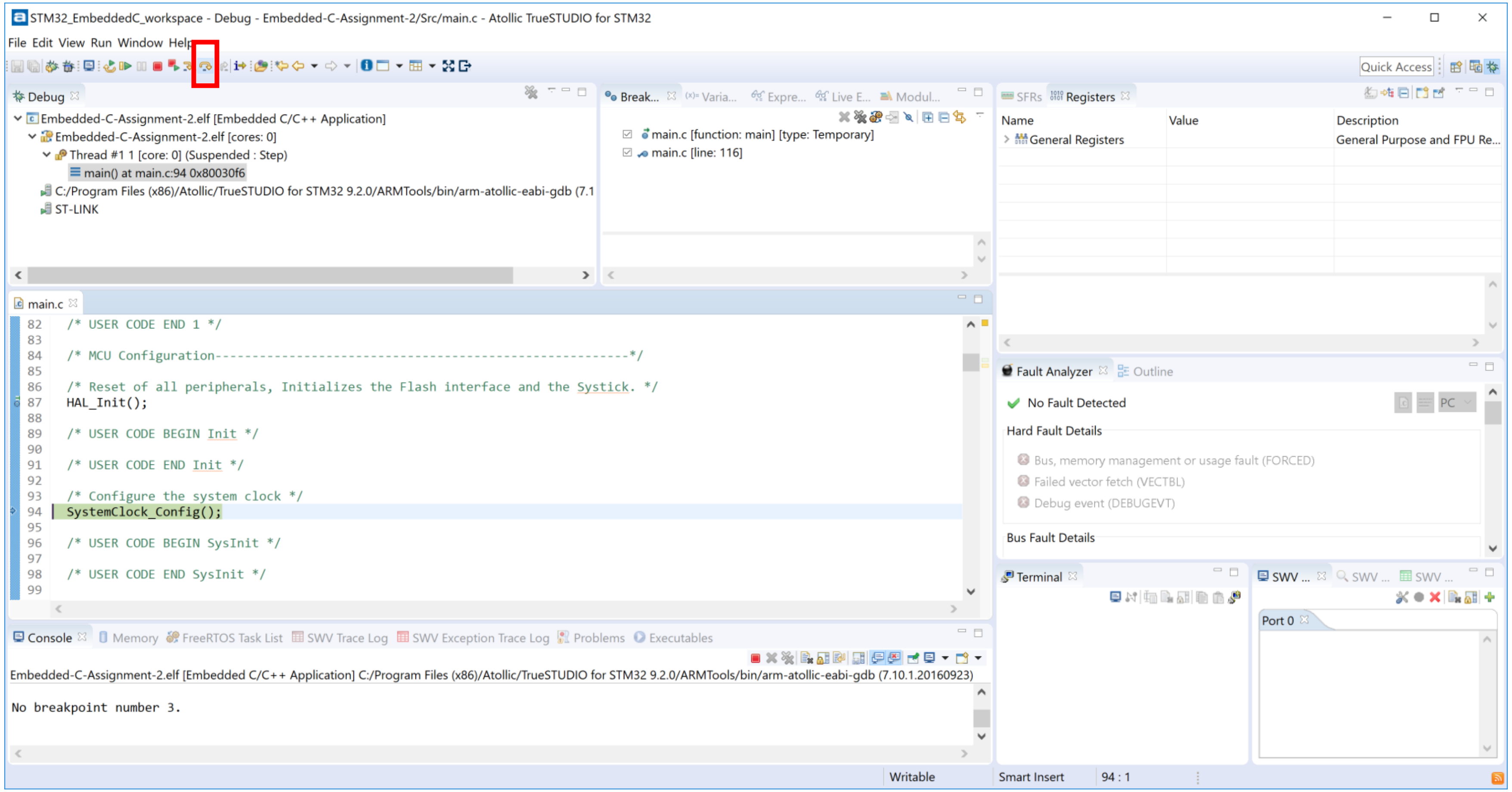
# Step 23. Run in Debug



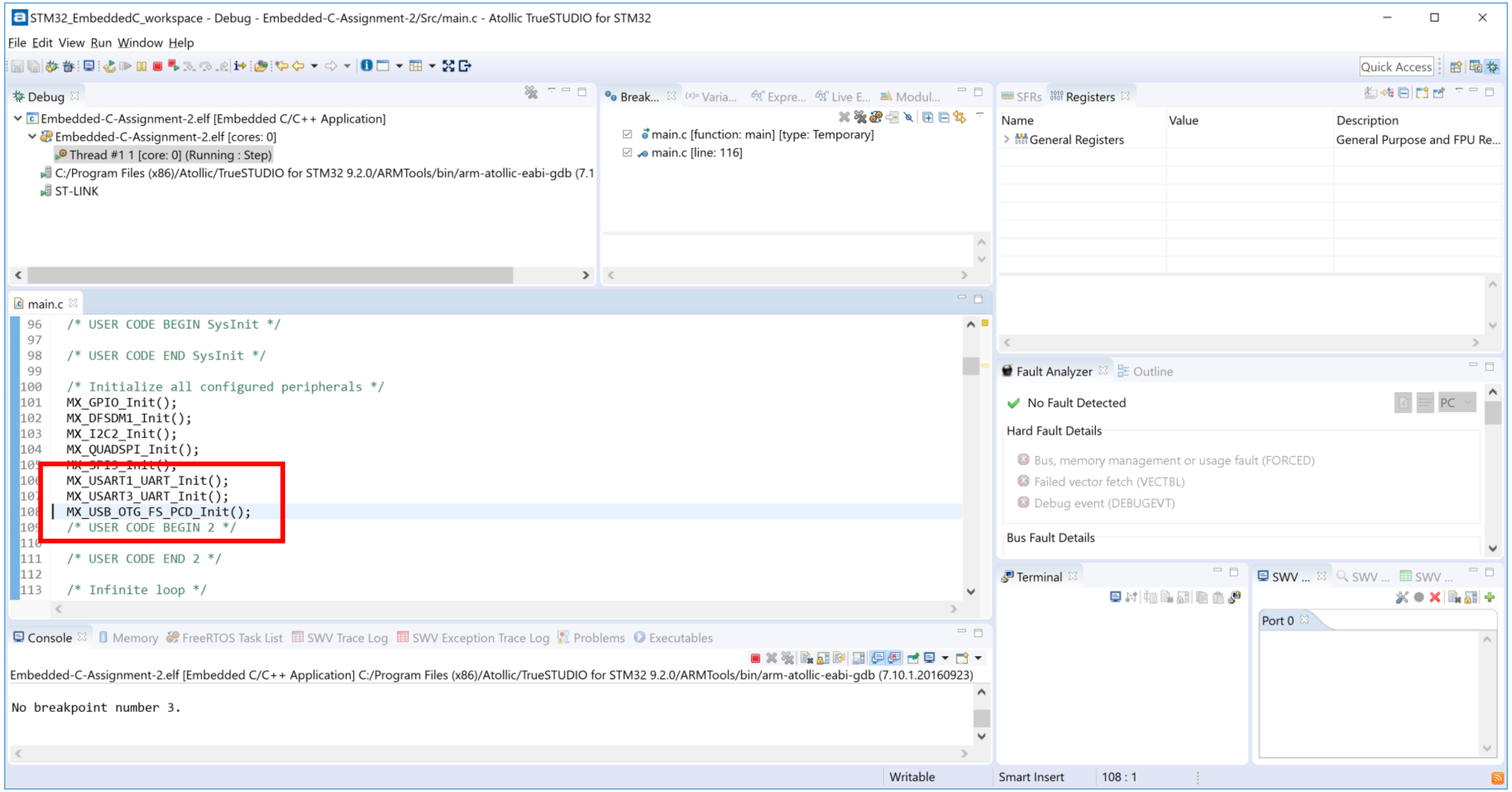
# Step 24. Hit Breakpoint



# Step 25. Click “Step Over”.



# Step 26. Click "Step Over". Repeat as needed. Mine "hung" at MX_USB_OTG_FS_PCD_Init()



# Step 27. Comment out MX_USB_OTG_FS_PCD_Init(), then rebuild

*main.c

```
  /* USER CODE BEGIN SysInit */

  /* USER CODE END SysInit */

  /* Initialize all configured peripherals */
  MX_GPIO_Init();
  MX_DFSDM1_Init();
  MX_I2C2_Init();
  MX_QUADSPI_Init();
  MX_SPI3_Init();
  MX_USART1_UART_Init();
  MX_USART3_UART_Init();
  //NOT USING USB - THIS FUNCTION HUNG MX_USB_OTG_FS_PCD_Init();
  /* USER CODE BEGIN 2 */

  /* USER CODE END 2 */

  /* Infinite loop */
  /* USER CODE BEGIN WHILE */

  uint32_t flash_size = LL_GetFlashSize();
  printf("flash_size: %lx\n", flash_size);

  uint32_t uid[3];
  uid[0] = LL_GetUID_Word0();
```

# Step 28. Build, Run, Use Debugger to set through code. Notice LED toggles on/off